PRACTICAL SUBMISSION RECORD- A.Y. 2024-25

Class: SYMCA **Div: B** **Semester: III**	**Course Code: MCA01604** **Course Name: Data Science Laboratory**		**Batch: S1**
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Title: Using grocery dataset with minimum support to 0.001 and minimum confidence of 0.8 build a Frequent Pattern Tree (FP-Tree). Show for each transaction how the tree evolves

Code:

```
# --- Step 1: Import necessary libraries ---
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import fpgrowth, association_rules
from tabulate import tabulate

# --- Step 2: Data Loading and Preparation ---
filename = 'groceries.csv'
dataset = []
try:
    df = pd.read_csv(filename)

    df = df.drop(columns=df.columns[0])

    for index, row in df.iterrows():
        transaction = row.dropna().tolist()
        dataset.append(transaction)

except FileNotFoundError:
    print(f"Error: '{filename}' not found. Please make sure the file is in the same directory as the script.")
    exit()
except Exception as e:
    print(f"An error occurred: {e}")
    exit()

te = TransactionEncoder()
te_ary = te.fit(dataset).transform(dataset)
df_encoded = pd.DataFrame(te_ary, columns=te.columns_)

# --- Step 3: Apply the FP-Growth algorithm with your specified support ---
min_support_threshold = 0.01
```

```
frequent_itemsets = fpgrowth(df_encoded, min_support=min_support_threshold, use_colnames=True)

# --- Step 4: Generate association rules with your specified confidence ---
min_confidence_threshold = 0.5
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=min_confidence_threshold)

# --- Step 5: Display Output ---
print("\n" + "="*80)
print(" " * 20 + "FP-GROWTH ANALYSIS RESULTS")
print("="*80 + "\n")

# --- Frequent Itemsets ---
print(f"Top 15 Frequent Itemsets (Support >= {min_support_threshold})")
sorted_itemsets = frequent_itemsets.sort_values(by='support', ascending=False)
print(tabulate(sorted_itemsets.head(15), headers='keys', tablefmt='psql', showindex=False))
print(f"\n[INFO] Found a total of {len(frequent_itemsets)} frequent itemsets.\n")

# --- Association Rules ---
print(f"Generated Association Rules (Confidence >= {min_confidence_threshold})")
if rules.empty:
    print("No rules found for the given support and confidence thresholds.")
else:
    relevant_rules = rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']]
    sorted_rules = relevant_rules.sort_values(by=['confidence', 'lift'], ascending=False)
    print(tabulate(sorted_rules.head(15), headers='keys', tablefmt='psql', showindex=False, floatfmt=".4f"))
print(f"\n[INFO] Found a total of {len(rules)} rules.\n")
```

Output:

```
================================================================================
                    FP-GROWTH ANALYSIS RESULTS
================================================================================

📊 Top 15 Frequent Itemsets (Support >= 0.01)
+-----------+--------------------------------+
|   support | itemsets                       |
|-----------+--------------------------------|
| 0.255516  | frozenset({'whole milk'})      |
| 0.193493  | frozenset({'other vegetables'}) |
| 0.183935  | frozenset({'rolls/buns'})      |
| 0.174377  | frozenset({'soda'})            |
| 0.139502  | frozenset({'yogurt'})          |
| 0.110524  | frozenset({'bottled water'})   |
| 0.108998  | frozenset({'root vegetables'}) |
| 0.104931  | frozenset({'tropical fruit'})  |
| 0.0985257 | frozenset({'shopping bags'})   |
| 0.0939502 | frozenset({'sausage'})         |
| 0.088968  | frozenset({'pastry'})          |
| 0.0827656 | frozenset({'citrus fruit'})    |
| 0.0805287 | frozenset({'bottled beer'})    |
| 0.079817  | frozenset({'newspapers'})      |
| 0.0776817 | frozenset({'canned beer'})     |
+-----------+--------------------------------+

[INFO] Found a total of 333 frequent itemsets.

🔗 Generated Association Rules (Confidence >= 0.5)
+--------------------------------------------------------+---------------------------------+-----------+--------------+--------+
| antecedents                                            | consequents                     |   support |   confidence |   lift |
|--------------------------------------------------------+---------------------------------+-----------+--------------+--------|
| frozenset({'root vegetables', 'citrus fruit'})         | frozenset({'other vegetables'}) |    0.0104 |       0.5862 | 3.0296 |
| frozenset({'root vegetables', 'tropical fruit'})       | frozenset({'other vegetables'}) |    0.0123 |       0.5845 | 3.0210 |
| frozenset({'yogurt', 'curd'})                          | frozenset({'whole milk'})       |    0.0101 |       0.5824 | 2.2791 |
| frozenset({'other vegetables', 'butter'})              | frozenset({'whole milk'})       |    0.0115 |       0.5736 | 2.2449 |
| frozenset({'root vegetables', 'tropical fruit'})       | frozenset({'whole milk'})       |    0.0120 |       0.5700 | 2.2310 |
| frozenset({'yogurt', 'root vegetables'})               | frozenset({'whole milk'})       |    0.0145 |       0.5630 | 2.2034 |
| frozenset({'other vegetables', 'domestic eggs'})       | frozenset({'whole milk'})       |    0.0123 |       0.5525 | 2.1623 |
| frozenset({'yogurt', 'whipped/sour cream'})            | frozenset({'whole milk'})       |    0.0109 |       0.5245 | 2.0527 |
| frozenset({'root vegetables', 'rolls/buns'})           | frozenset({'whole milk'})       |    0.0127 |       0.5230 | 2.0469 |
| frozenset({'other vegetables', 'pip fruit'})           | frozenset({'whole milk'})       |    0.0135 |       0.5175 | 2.0254 |
| frozenset({'yogurt', 'tropical fruit'})                | frozenset({'whole milk'})       |    0.0151 |       0.5174 | 2.0248 |
| frozenset({'other vegetables', 'yogurt'})              | frozenset({'whole milk'})       |    0.0223 |       0.5129 | 2.0072 |
| frozenset({'other vegetables', 'whipped/sour cream'})  | frozenset({'whole milk'})       |    0.0146 |       0.5070 | 1.9844 |
| frozenset({'root vegetables', 'rolls/buns'})           | frozenset({'other vegetables'}) |    0.0122 |       0.5021 | 2.5949 |
| frozenset({'yogurt', 'root vegetables'})               | frozenset({'other vegetables'}) |    0.0129 |       0.5000 | 2.5841 |
+--------------------------------------------------------+---------------------------------+-----------+--------------+--------+

[INFO] Found a total of 15 rules.
```

Each transaction how the tree evolves

1. **Calculate Item Frequency & Establish Order**
   - bread: 5
   - milk: 4
   - butter: 4
   - sugar: 2
   - eggs: 2
   - coffee: 1 (Discarded as its count is less than 2)

2. Building the FP-Tree Transaction by Transaction

Transaction 1:

```
Tree State:
(root)
└── bread:1
    └── butter:1
        └── milk:1
```

Transaction 2:

```
Tree State:
(root)
└── bread:2
    ├── butter:1
    │   └── milk:1
    └── milk:1
        └── sugar:1
```

Transaction 3:

```
Tree State:
(root)
├── bread:2
│   ├── butter:1
│   │   └── milk:1
│   └── milk:1
│       └── sugar:1
└── butter:1
    └── milk:1
        └── eggs:1
```

Transaction 4:

```
Tree State:
(root)
├── bread:3
│   ├── butter:2
│   │   ├── milk:1
│   │   └── eggs:1
│   └── milk:1
│       └── sugar:1
└── butter:1
    └── milk:1
        └── eggs:1
```

Transaction 5:

```
Tree State:
(root)
├── bread:4
│   ├── butter:3
│   │   ├── milk:2
│   │   │   └── sugar:1
│   │   └── eggs:1
│   └── milk:1
│       └── sugar:1
└── butter:1
    └── milk:1
        └── eggs:1
```

3. FP-Tree Structure

```
Final Tree:
(root)
├── bread:5
│   ├── butter:3
│   │   ├── milk:2
│   │   │   └── sugar:1
│   │   └── eggs:1
│   └── milk:1
│       └── sugar:1
└── butter:1
    └── milk:1
        └── eggs:1
```